

CLIENT

React Frontend (Vite · port 80)

- HomePage
- AutonomousAgent
- Results
- Report
- Sessions
- Workspaces
- PatientSelection
- Settings
- Toast (SSE)
- NavDock

HTTP / SSE

REVERSE PROXY

nginx

proxy_pass /api/ → orchestrator:5001 · HTTP/1.1 · proxy_buffering off

HTTP/1.1

ORCHESTRATOR · FASTAPI · PORT 5001

agenticAgent

Gemini / Ollama loop
multi-turn function calls
sessionContext
gemini_client / ollama_client

task_runner

background thread pool
run_inference
_explain_error (LLM)
stderr log capture

watcher_service

watchdog filesystem
per-directory polling
triggers agent on new files
console SSE stream

tool_registry

MCP client manager · AsyncExitStack per server · session.list_tools / call_tool

backend.py — REST routes + SSE

/api/query · /api/events · /api/tasks · /api/sessions · /api/upload · /api/diagnostics · auth

SQLITE · DATA/HEALTH.DB

users sessions session_context conversation_messages uploaded_files background_tasks watched_directories user_tool_settings

EXTERNAL LLM

Gemini API

google.generativeai

Ollama

local inference · REST API

MCP SERVER

monai

stdio

analyze_image
list_models
download_model
get_monai_info
run_inference (BG)

MCP SERVER

radlex

stdio

list_subspecialties
find_templates
get_template_schema
generate_report

MCP SERVER

utils

stdio

parse_dicom
parse_dicom_directory
write_file
list_directory
find_files

MCP SERVER

cellpose

stdio

segment_cells_2d
segment_cells_3d
save_overlay

MCP SERVER

ipath

stdio

fetch_thumbnail
get_slide_dimensions
scale_roi_to_slide
fetch_roi
fetch_dicom_instance

MCP SERVER

skills

stdio

read_skill_file
execute_script
read_references
read_asset

MCP SERVER

fhir

HTTP

FHIR patient data
localhost:8000/mcp

MONAI bundles

model weights ·
HuggingFace Hub

ipath.bmd-

software.com

Dicoogle · DICOMweb · WSI